

Segmentation Free Recognition From Telugu Palm Leaves

K. Jaswanth¹
Artificial Intelligence
Amrita School of Computing
BL.SC.U4AIE24127
Bengaluru, India

G. Nivas²
Artificial Intelligence
Amrita School of Computing
BL.SC.U4AIE24118
Bengaluru, India

K. Manish³
Artificial Intelligence
Amrita School of Computing
BL.SC.U4AIE24129
Bengaluru, India

I. INTRODUCTION

It analyzes data using the Machine Learning approach and making predictions. In this project, the machine learning methods are used to process a dataset with DCT features of malware samples.

To approximate the relationship between input features and the target label and test the performance of the predictions, we use Linear Regression to model the relationship, and calculate a selection of metrics, including MSE, RMSE, MAPE, and R² score. K-Means Clustering is also used to cluster the similar data points and the quality of the clustering is determined through Silhouette Score, Calinski-Harabasz Score and Davies-Bouldin Index. The method of determining the number of optimum clusters is through the Elbow Method.

II. METHODOLOGY

A. Dataset Loading

The dataset DCT-mal.csv is loaded using the Pandas library. This dataset contains multiple numerical features and a target column labeled LABEL. The features are used as input variables (X) and the label is used as the output variable (y).

B. Data Splitting

The dataset is divided into training and testing sets using the train-test-split() function. Around 80 of the data is used for training the model and 20 is used for testing. This helps in evaluating how well the model performs on unseen data.

C. Linear Regression using a Single Feature

A Linear Regression model is first trained using only one attribute (feature column 0). This step helps to observe how well a single feature can predict the target label.

D. Linear Regression using All Features

Another Linear Regression model is trained using all available features in the dataset. This allows the model to learn from more information and usually improves prediction performance compared to using a single feature.

E. Regression Performance Evaluation

The regression models are evaluated using different metrics such as:

Mean Squared Error (MSE): Measures the average squared difference between actual and predicted values.

Root Mean Squared Error (RMSE): Square root of MSE, representing prediction error in the same unit as the data.

Mean Absolute Percentage Error (MAPE): Shows the percentage error between predicted and actual values.

R² Score: Indicates how well the model explains the variance in the data.

F. K-Means Clustering

K-Means clustering is applied to group similar data points into two clusters (k = 2) based on their feature similarity. It is an unsupervised learning technique used to identify hidden patterns in the dataset.

G. Clustering Evaluation Metrics

The quality of clustering is evaluated using the following metrics:

Silhouette Score: Measures how similar a data point is to its own cluster compared to other clusters.

Calinski-Harabasz Score: Evaluates cluster separation and compactness.

Davies-Bouldin Index: Measures the average similarity between clusters (lower values indicate better clustering).

H. Evaluation of Different K Values

The clustering process is repeated for different k values (2 to 9) to analyze how the clustering quality changes with different numbers of clusters.

I. Elbow Method for Optimal Clusters

The Elbow Method is used to determine the optimal number of clusters. Distortion (inertia) values are calculated for different k values, and a graph is plotted to identify the point where the reduction in distortion starts to slow down. This point represents the optimal number of clusters.

III. RESULTS

For One Feature		
	Train	Test
MSE	887329.2571577085	760196.9424258426
RMSE	941.9815588203988	871.8927356193781
MAPE	9226.495633558789	7900.868528270537
R2	0.000983255154679341	0.17661880372909855

TABLE I

PERFORMANCE METRICS OF THE LINEAR REGRESSION MODEL USING A SINGLE FEATURE.

For All Features		
	Train	Test
MSE	658401.276219931	760196.9424258426
RMSE	811.4192974165275	871.8927356193781
MAPE	7052.784401768195	7900.868528270537
R2	0.2587262344102613	0.17661880372909855

TABLE II

PERFORMANCE METRICS OF THE LINEAR REGRESSION MODEL USING ALL FEATURES.

Clustering Evaluation Metrics	Score
Silhouette Score	0.044180784877168665
Calinski Harabasz Score	256.977229178106
Davies Bouldin Index	4.461685342742857

TABLE III

EVALUATION METRICS OF THE K-MEANS CLUSTERING MODEL.

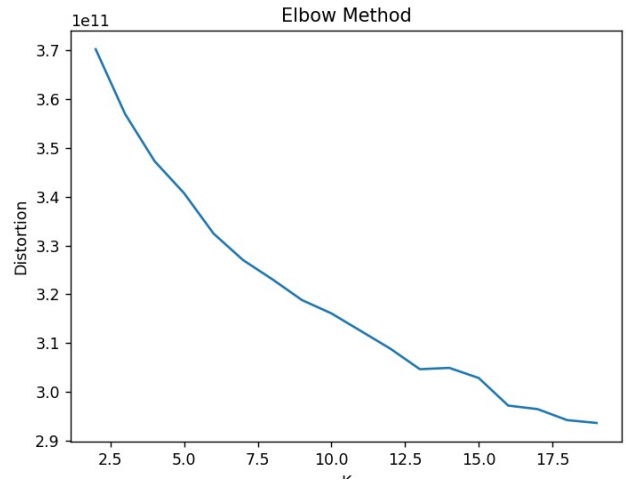


Fig. 2. Linear regression prediction results using all features.

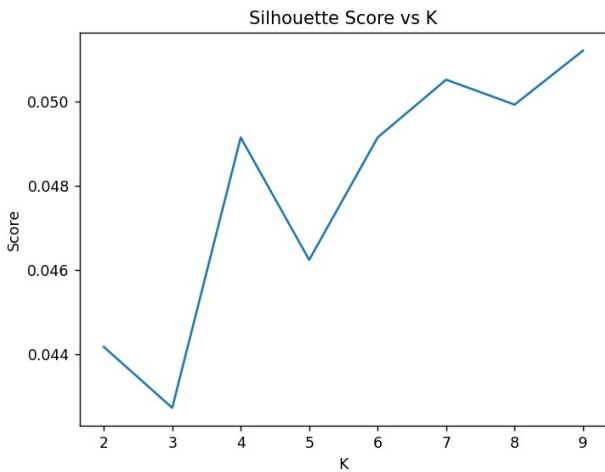


Fig. 1. Linear regression prediction results using one feature.